

Module2. Underlying Mechanisms of Deep Learning

Intelligent Video Analytics with Deep Learning

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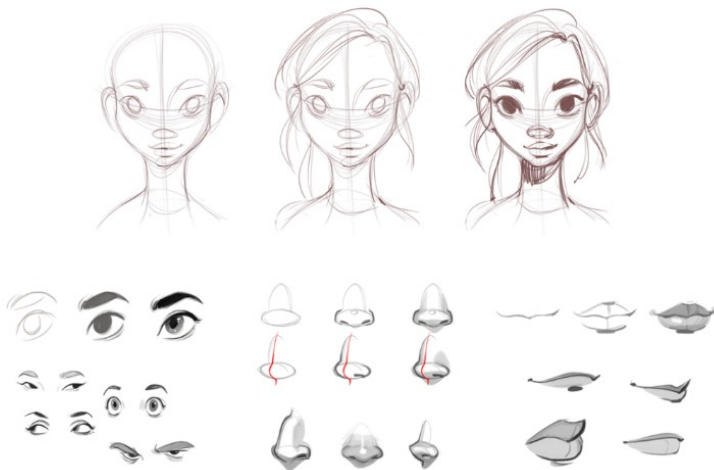
Note. This content mainly refers to the summer session of KAIST organized by Jiyong Park(2018)

Module2. Underlying Mechanisms of Deep Learning

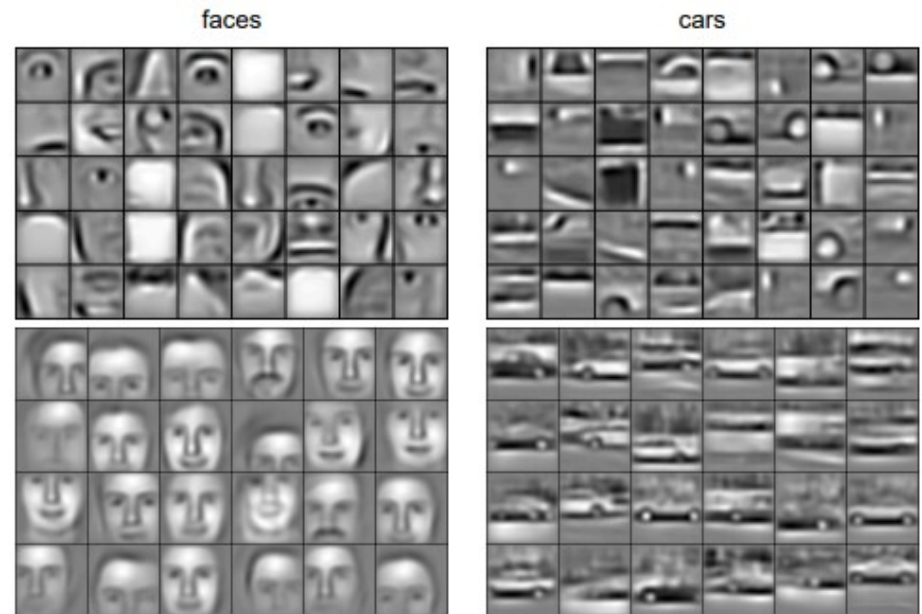
Intuitive Understanding of Deep Learning

- What is deep learning?
 - Deep learning is “**Representation Learning**” to learn and discover how to represent features of data. (Here, representation means a machine-understandable format)

How to draw faces?



How to represent faces?



Intuitive Understanding of Deep Learning

- Why important to well-represent the features of data?
- What needs to well-represent the features of data?
- Why is deep learning outstanding at representation learning?

Intuitive Understanding of Deep Learning

- Why important to well-represent the features of data? —————→
- What needs to well-represent the features of data? —————→
- Why is deep learning outstanding at representation learning? —————→

Ch1. Representational Learning

Ch2. Artificial Neural Networks

Ch3. Deep Learning Algorithms

Ch1. Representational Learning

What is the meaning of Well-Representation of the Data Feature?

- First, let's look at an example of what a Representation is:



- How can you distinguish the objects?

What is the meaning of Well-Representation of the Data Feature?

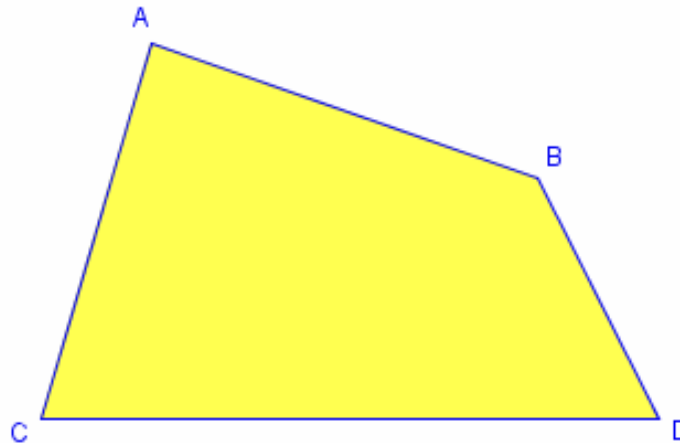
- First, let's look at an example of what a Representation is:



- If you try to solve the problem of Classify the following shape, the number of corners can be used as a Representation
 - Square: 4 corner
 - Triangle: 3 corner
 - Circle: 0 corner

What is the meaning of Well-Representation of the Data Feature?

- But how do we cope if this shape comes in?



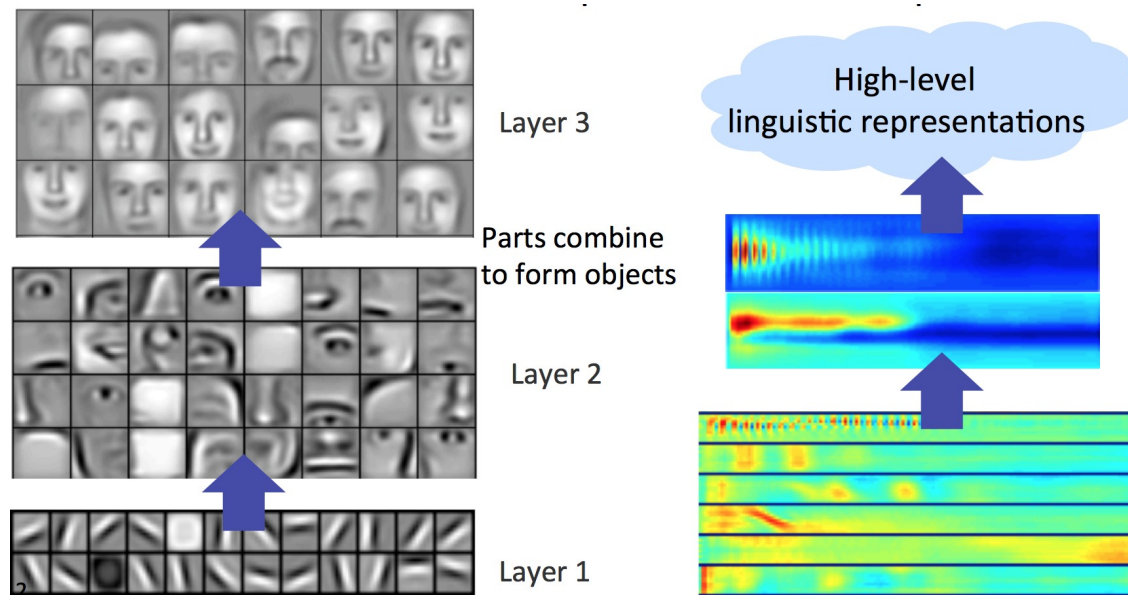
- You need great expertise in good representation, and Deep Learning automatically learns this representation

Why Should We Care about Learning Representation?

- The performance of machine learning methods is heavily dependent on the choice of data representation
- Conventional machine-learning techniques were limited in their ability to process natural data in their raw form.

Why Should We Care about Learning Representation?

- Deep-learning methods are representation-learning methods with multiple levels of representation from a concrete level to slightly more abstract level.

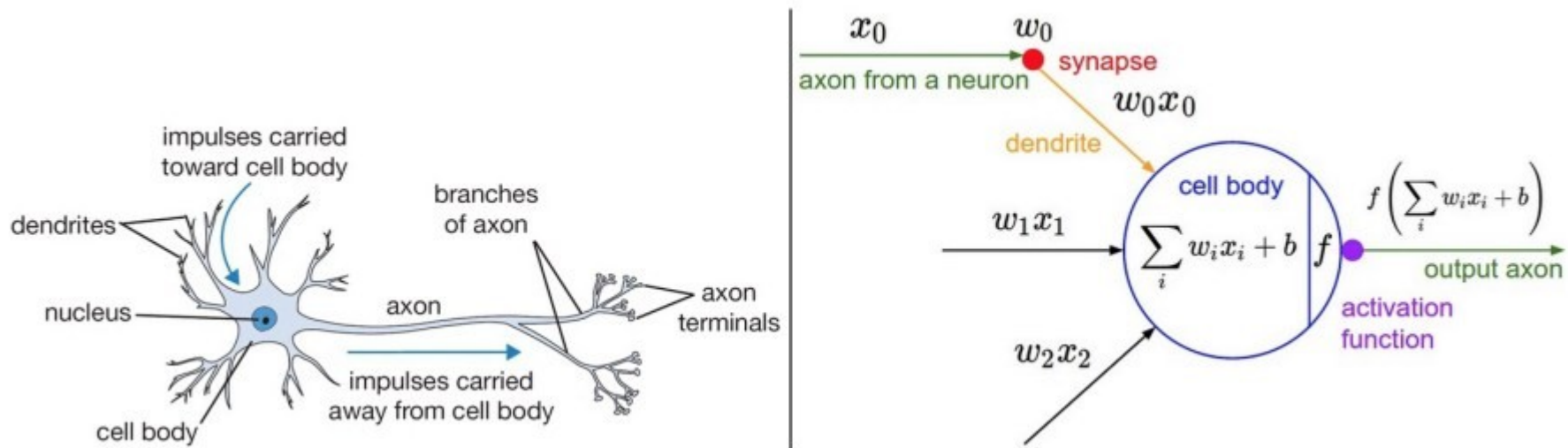


- Very complex functions can be learned by learning representation using Deep-learning

Ch2. Artificial Neural Networks

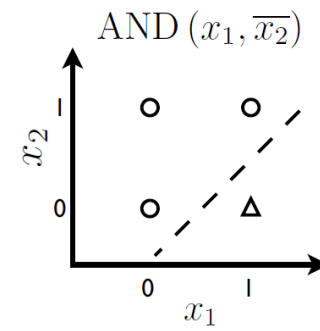
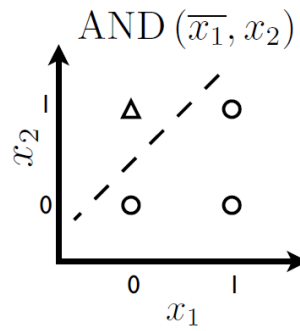
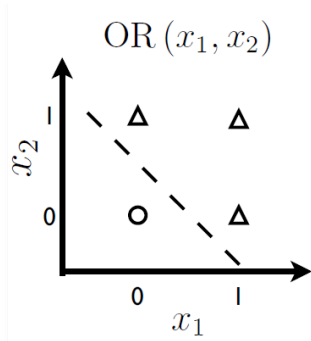
Neural Networks

- Artificial neural network mimics the human brain
 - While neural network-based algorithms for classification or regression may be useful for the purpose of artificial intelligence (AI), neural network itself has nothing to do with.
 - Single neuron is activated (=1) or not (=0).

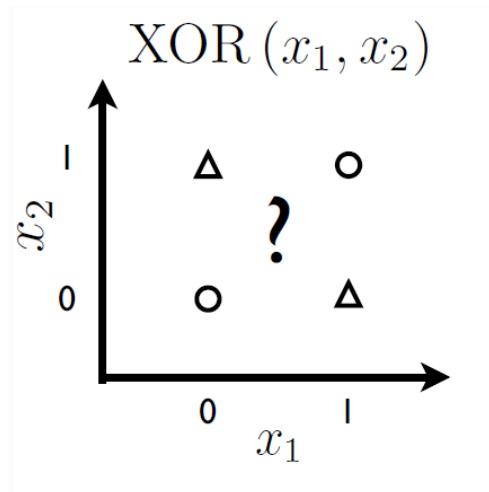


Capacity of Neural Networks

- Linearly separable problems

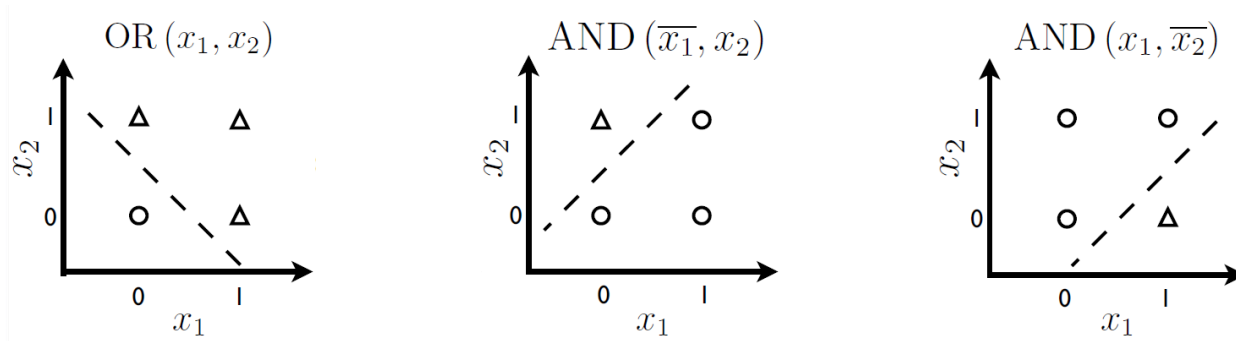


- Non-linearly separable problems

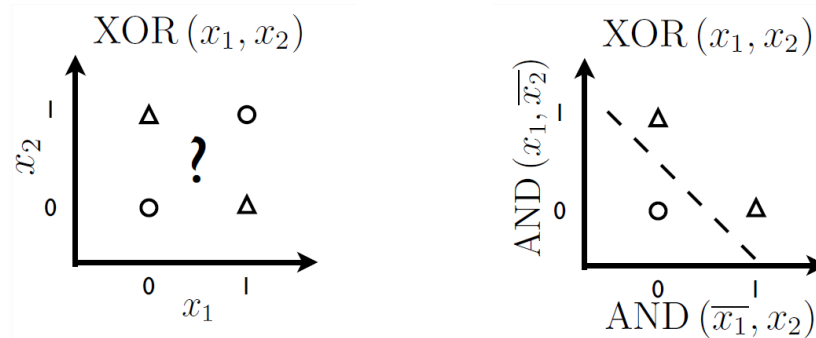


Capacity of Neural Networks

- Artificial neural networks can solve non-linearly separable problems.
 - That is, neural networks can represent more complex features!
- Linearly separable problems



- Non-linearly separable problems

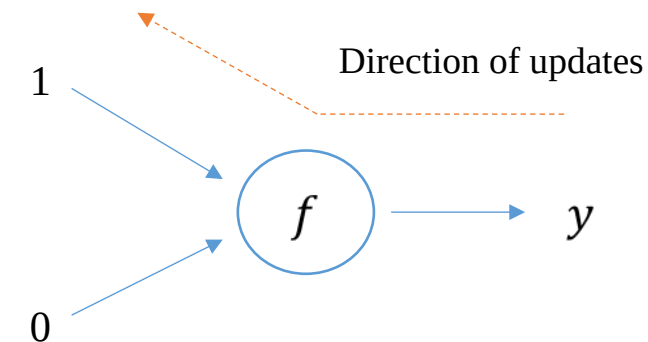


Multi-layer neural networks can solve it by transforming the input in a better representation.

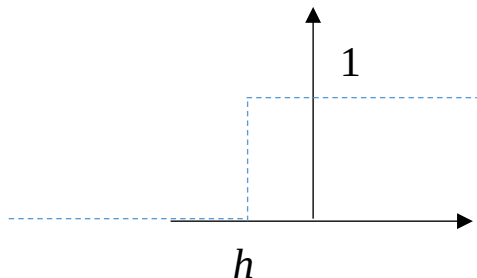
Learning Algorithm of Neural Networks

- Back-propagation

= Learning parameters to minimize the error from the true value in the reverse way



Activation function
 $y = f(w_1 * 1 + w_0 * 0)$



(1) If the model outputs 1 and the true value is 0, how should you modify the parameters (w_1, w_0, h)?

(2) If the model outputs 0 and the true value is 1, how should you modify the parameters (w_1, w_0, h)?

(3) Until when? Minimizing *the error (loss) function*

Ch3. Deep Learning Algorithms

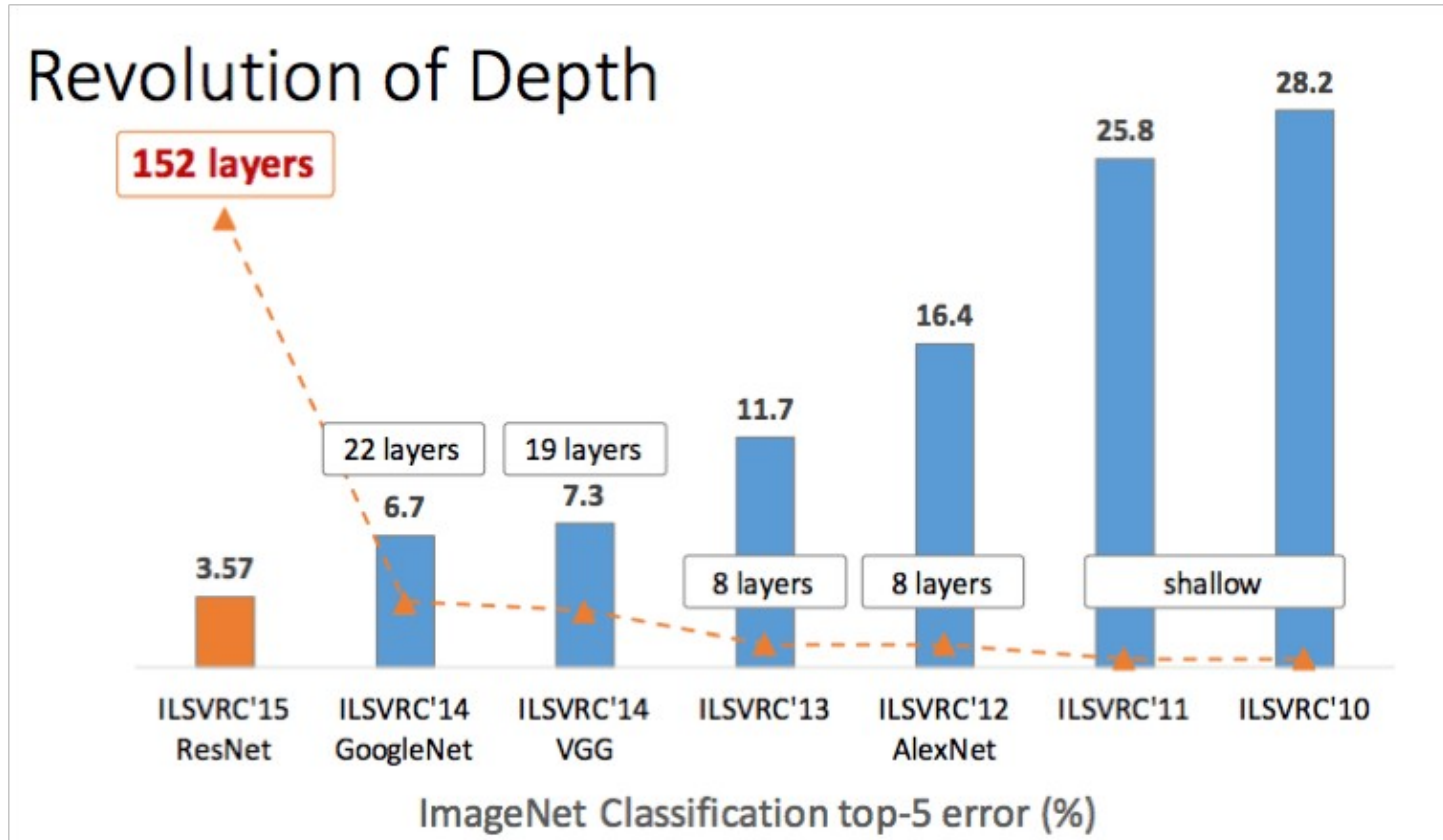
Representation Learning + Deep Architecture

- “Deep Learning can be considered as special case of representation learning algorithms which learn representations of the data in a Deep Architecture with **multiple levels of representations**.” (Najafabadi et al. 2015, p. 5)
- Recent deep learning algorithms are mostly based on neural networks.
 - Neural network + Deep architecture
 - = Feed-forward networks with many hidden layers
 - = Multi-layer perceptron (MLP)
 - = Deep neural network (DNN)
 - More specialized algorithms: CNN, RNN, LSTM, etc.

Najafabadi, M. M., Villanustre, F., Khoshgoftaar, T. M., Seliya, N., Wald, R., & Muharemagic, E. (2015). Deep Learning Applications and Challenges in Big Data Analytics. *Journal of Big Data*, 2(1), 1.

Representation Learning + Deep Architecture

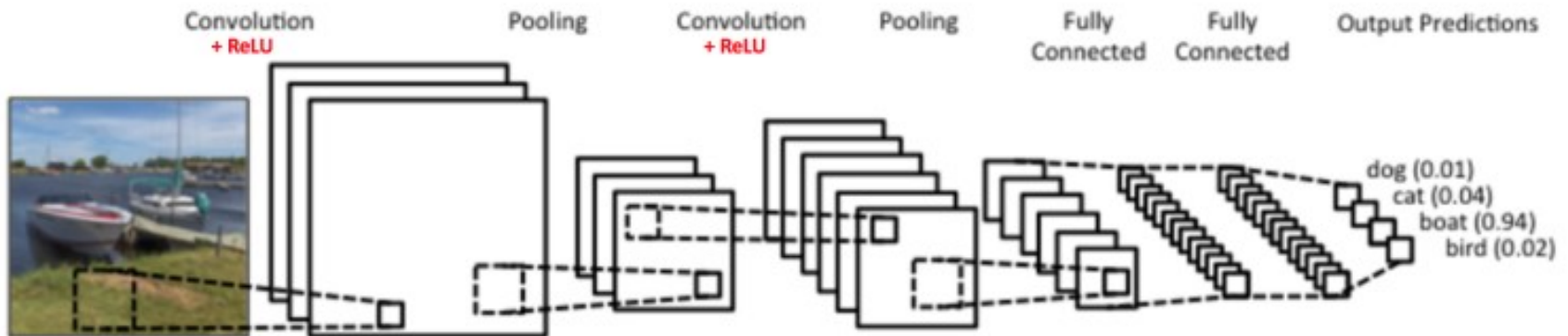
- Then, how deep?



Convolutional Neural Network

What is Convolutional Neural Network (CNN)?

- A typical example of CNN



- Inputs (2-dimension $\times$ channels) $\rightarrow$ [Convolution $\rightarrow$ ReLu $\rightarrow$ Pooling] $\rightarrow$... $\rightarrow$ [Convolution $\rightarrow$ ReLu $\rightarrow$ Pooling] $\rightarrow$ Fully connected layer $\rightarrow$ Output prediction (Multi-class classification)

Example: 1D Convolution

- How can we apply convolution on 1D time series data?
- 1D Convolution of discrete time signals
- Example

- Kernel:

-1	1
----	---

- Time series data:

0	0	0	1	0	1	1	1
---	---	---	---	---	---	---	---

Example: 1D Convolution

- Example

- $N = 0: -1*0 + 1*0 = 0$

-1	1
----	---

0	0	0	1	0	1	1	1
---	---	---	---	---	---	---	---

- $N = 1: -1*0 + 1*0 = 0$

-1	1
----	---

0	0	0	1	0	1	1	1
---	---	---	---	---	---	---	---

- $N = 2: -1*0 + 1*1 = 1$

-1	1
----	---

0	0	0	1	0	1	1	1
---	---	---	---	---	---	---	---

Example: 2D Convolution

- Convolution

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Image data

1	0	1
0	1	0
1	0	1

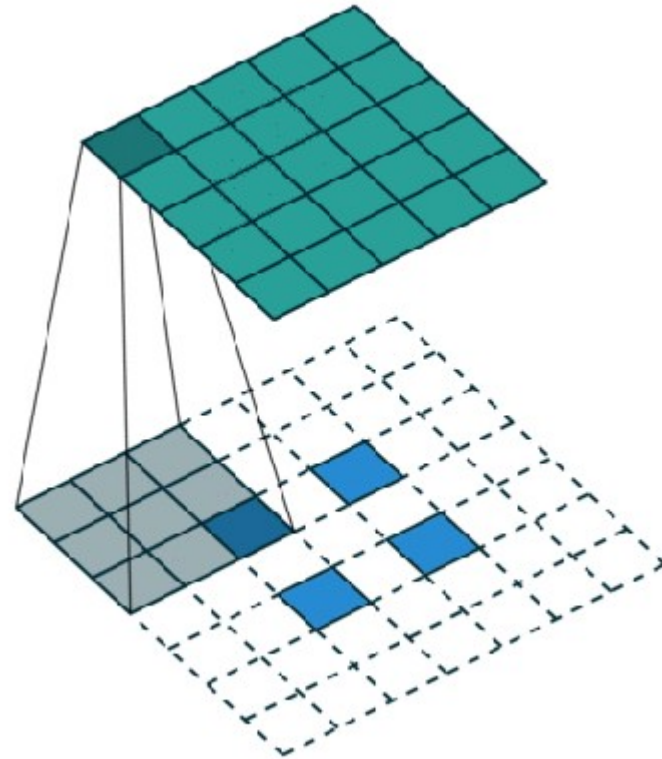
Convolution filter

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

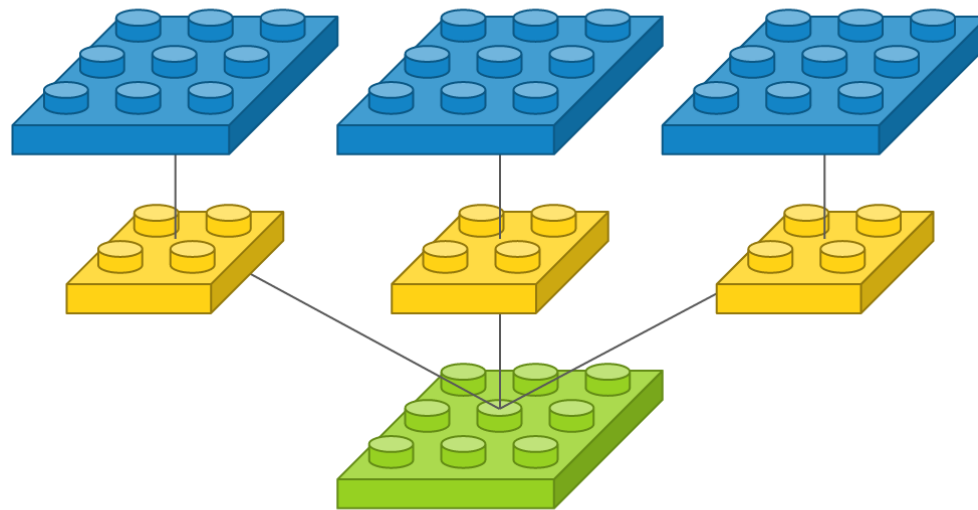
Image

4		

Convolved
Feature

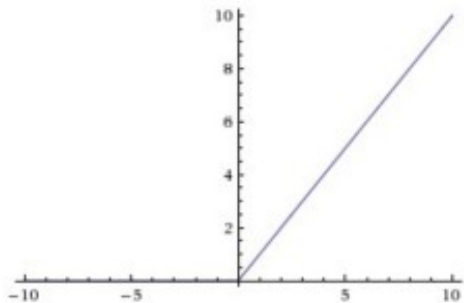


Applying 2D Multiple Kernel

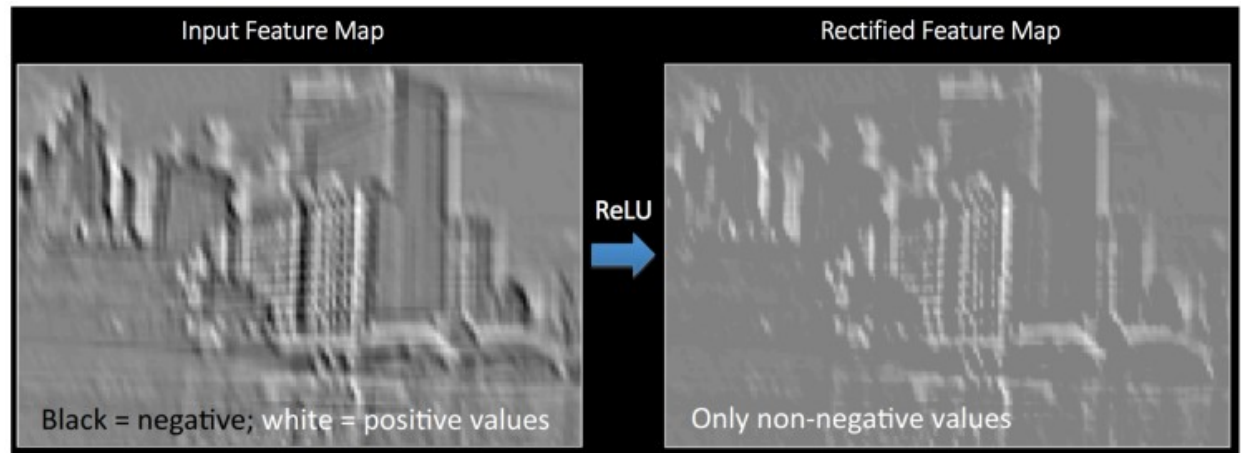


ReLU

- Introducing non-linearity (ReLU)
 - The purpose of ReLU is to introduce non-linearity in CNN.
 - Other non-linear functions such as tanh or sigmoid can also be used, but ReLU has been found to perform better in most situations.



Output = $\text{Max}(\text{zero}, \text{Input})$



Source: An Intuitive Explanation of Convolutional Neural Networks (Data Science Blog)
<https://ujjwalkarn.me/2016/08/11/intuitive-explanation-convnets/>

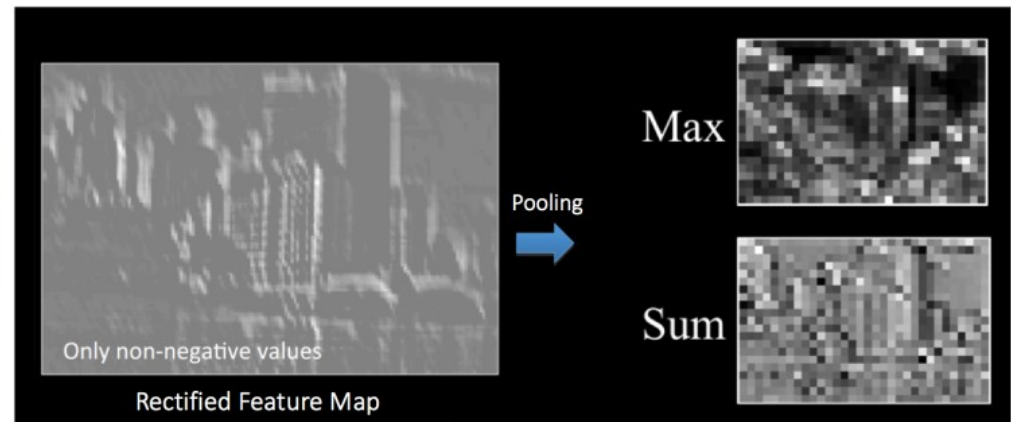
Max Pooling

- Reducing the dimensionality of each feature map (Pooling)
 - The purpose of pooling is to reduce the number of parameters and computations in the network, therefore, controlling overfitting.
 - Another purpose is to make the network invariant to small transformations, distortions and translations in the input image.
 - Pooling can be of different types: Max, Average, Sum, etc.

12	20	30	0
8	12	2	0
34	70	37	4
112	100	25	12

2×2 Max-Pool

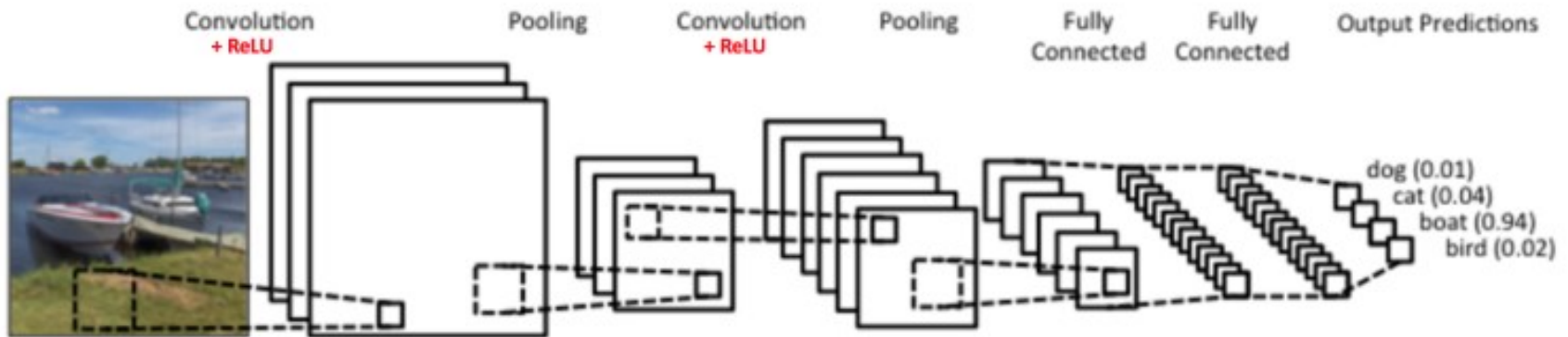
20	30
112	37



Source: An Intuitive Explanation of Convolutional Neural Networks (Data Science Blog)
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What is Convolutional Neural Network (CNN)?

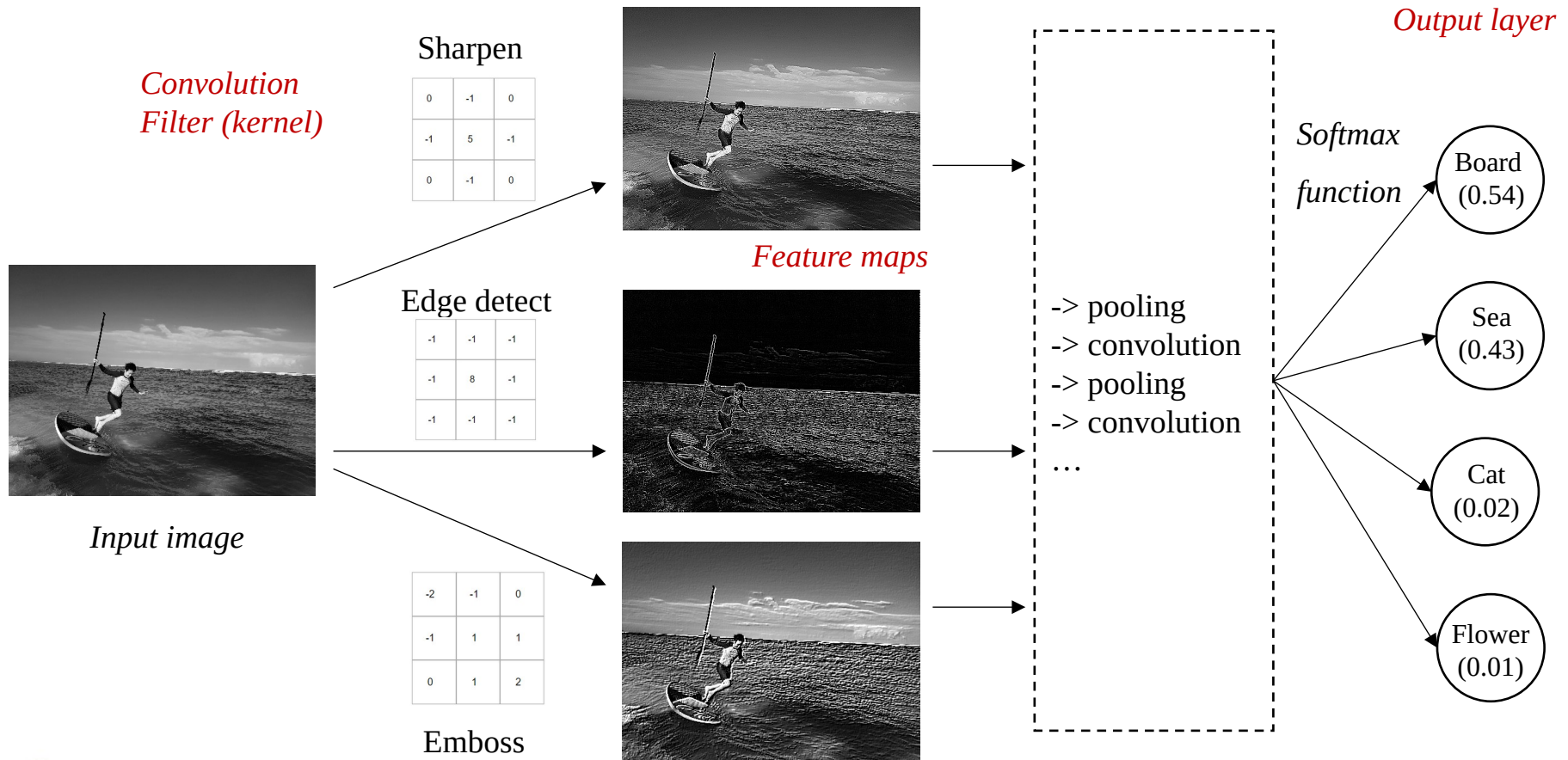
- A typical example of CNN



- Inputs (2-dimension $\times$ channels) $\rightarrow$ [Convolution $\rightarrow$ ReLu $\rightarrow$ Pooling] $\rightarrow$... $\rightarrow$ [Convolution $\rightarrow$ ReLu $\rightarrow$ Pooling] $\rightarrow$ Fully connected layer $\rightarrow$ Output prediction (Multi-class classification)

Convolutional Neural Network (CNN)

- CNN automatically learns the best filters to extract the feature maps for image classifications through back propagation.



Convolutional Neural Network (CNN)

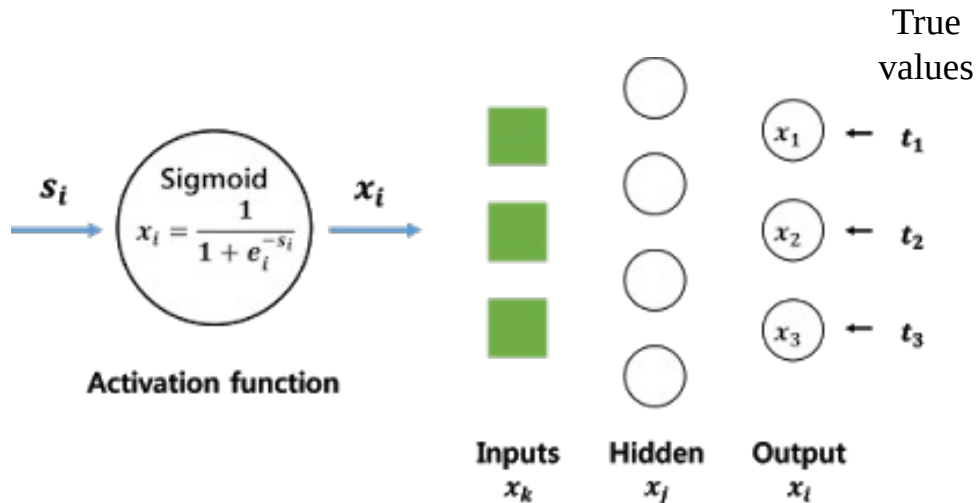
- CNN has been considered as a basic deep learning algorithm. Why is CNN superior at feature representations?
 - Effectively reducing computational complexity (Convolution filter)
 - Hierarchical feature representation (multiple convolution layers)
 - Non-linearity and less overfitting (Rectified Linear Unit (ReLU) and Pooling)

Thank you □

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(Appendix) Learning Algorithm of Neural Networks

- Gradient descent method (Back-propagation)
 - = Learning parameters to minimize the error from the true value in the systematic, reverse way even for multiple hidden layers



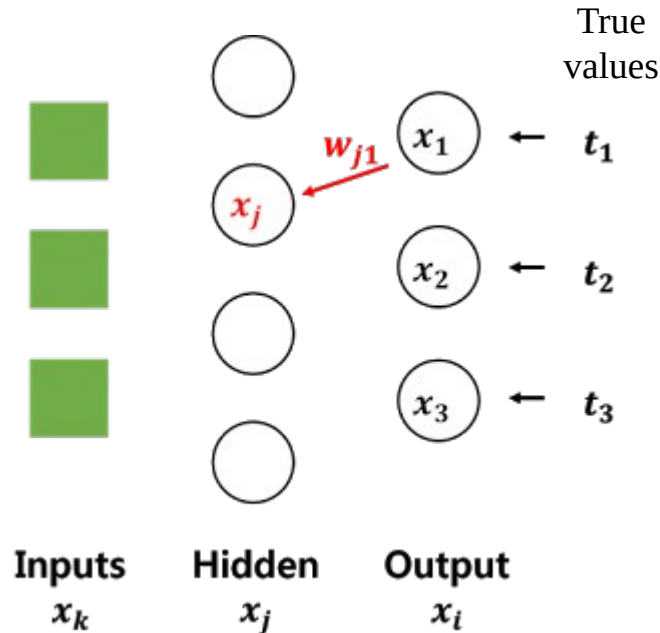
The goal is to minimize the following error (loss) function,

$$E = - \sum_{i=1}^3 [t_i \log(x_i) + (1 - t_i) \log(1 - x_i)]$$

Source: Fully Understanding Backpropagation with Examples (Jaejun Yoo's Playground) <http://jaejunyoo.blogspot.com/2017/01/backpropagation.html>

(Appendix) Learning Algorithm of Neural Networks

- Gradient descent method (Back-propagation)
 - = Learning parameters to minimize the error from the true value in the systematic, reverse way even for multiple hidden layers



Firstly, update the parameters between output and hidden layers.

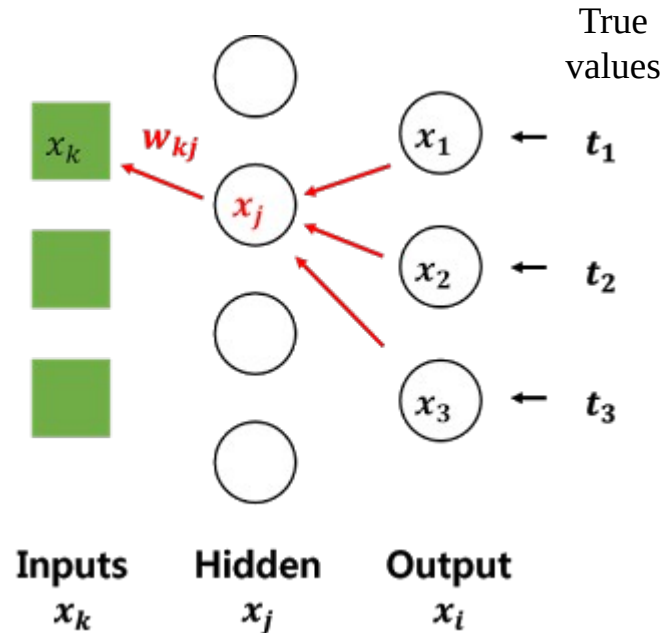
$$\frac{\partial E}{\partial w_{j1}} = \frac{\partial E}{\partial x_i} \frac{\partial x_i}{\partial s_i} \frac{\partial s_i}{\partial w_{ji}} \quad (\text{Chain rule})$$

$$\therefore \frac{\partial E}{\partial w_{j1}} = (x_i - t_i)x_j$$

Source: Fully Understanding Backpropagation with Examples (Jaejun Yoo's Playground) <http://jaejunyoo.blogspot.com/2017/01/backpropagation.html>

(Appendix) Learning Algorithm of Neural Networks

- Gradient descent method (Back-propagation)
 - = Learning parameters to minimize the error from the true value in the systematic, reverse way even for multiple hidden layers



Secondly, update the parameters between hidden and input layers.

$$\frac{\partial E}{\partial w_{kj}} = \frac{\partial E}{\partial s_j} \frac{\partial s_j}{\partial w_{kj}}$$

$$\frac{\partial E}{\partial w_{kj}} = \sum_{i=1}^3 \frac{\partial E}{\partial s_i} \frac{\partial s_i}{\partial x_j} \frac{\partial x_j}{\partial s_j} \times \frac{\partial s_j}{\partial w_{kj}}$$

$$\therefore \frac{\partial E}{\partial w_{kj}} = \sum_{i=1}^3 (x_i - t_i) w_{ji} (x_j (1 - x_j)) \times x_k$$

Source: Fully Understanding Backpropagation with Examples (Jaejun Yoo's Playground) <http://jaejunyoo.blogspot.com/2017/01/backpropagation.html>